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RNA_Seq_Analysis

mv conA_rep1.fq.gz conA_rep1.fq
mv conA_rep2.fq.gz conA_rep2.fq
mv conB_rep1.fq.gz conB_rep1.fq
mv conB_rep2.fq.gz conB_rep2.fq
file * .fq

Fastqc Analysis
fastqc -o fastqc_results \
conA_rep1.fq \
conA_rep2.fq \
conB_rep1.fq \
conB_rep2.fq



index reference genome

mkdir -p bowtie2_index
bowtie2-build GCF_000146045.2_R64_genomic.fna S288C

Alignment - We have to do it for all the four files

bowtie2 \
-x bowtie2_index/S288C \
-U conA_rep1.fq \
-S conA_rep1.sam

```
(bio-env) (base) ibab@IBAB-MSc-Comp11:~/bio-env$ bowtie2 \
-x S288C \
-U conA_rep1.fq \
-S conA_rep1.sam
[WARNING] Failed to launch x86-64-v3 version, staying with default
[WARNING] Failed to launch x86-64-v3 version, staying with default
800000 reads; of these:
  800000 (100.00%) were unpaired; of these:
    24266 (3.03%) aligned 0 times
    663604 (82.95%) aligned exactly 1 time
    112130 (14.02%) aligned >1 times
96.97% overall alignment rate
(bio-env) (base) ibab@IBAB-MSc-Comp11:~/bio-env$
```

We have to oad the Rsubread package for RNA-seq read alignment and feature counting

---

```{r}
library(Rsubread)
setwd("/home/ibab/bio-env")
gtf_file <- "GCF_000146045.2_R64_genomic.gtf"

out_FeatCount_conA_rep1 <- featureCounts(
  files = "conA_rep1.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)

out_FeatCount_conA_rep2 <- featureCounts(
```

```
  files = "conA_rep2.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)

out_FeatCount_conB_rep1 <- featureCounts(
  files = "conB_rep1.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)

out_FeatCount_conB_rep2 <- featureCounts(
  files = "conB_rep2.sam",
  annot.ext = gtf_file,
  isGTFAnnotationFile = TRUE,
  GTF.featureType = "exon",
  GTF.attrType = "gene_id"
)
```

We have to extract the gene-level count tables from each featureCounts result

```
out_FeatCount_conA_rep1_count = data.frame(out_FeatCount_conA_rep1$counts)
out_FeatCount_conA_rep2_count = data.frame(out_FeatCount_conA_rep2$counts)
out_FeatCount_conB_rep1_count = data.frame(out_FeatCount_conB_rep1$counts)
out_FeatCount_conB_rep2_count = data.frame(out_FeatCount_conB_rep2$counts)
```

We have to combine the four count tables into one count matrix

```
count_matrix <- cbind(
out_FeatCount_conA_rep1_count,
out_FeatCount_conA_rep2_count,
out_FeatCount_conB_rep1_count,
out_FeatCount_conB_rep2_count
```

```
)
write.table(count_matrix, file = "count_matrix.txt",
sep="\t")
```

```{r}
library(edgeR)
## Loading required package: limma
group=c(rep("conA",2),rep("conB",2))
```

Notice:

All remaining RStudio bioinformatic analysis procedures—including differential expression testing (edgeR), tagwise dispersion estimation, TMM normalization factor calculations, and statistical summaries, have been successfully executed, validated, and saved as “Debarshi_13_8_part2.pdf”